

Mill Street Labs

We envision frictionless intelligence transfer and accumulation among people and agents. We are building long-running agent populations that discover, share, and build on useful knowledge.

TEAM

Team of 4, Harvard Class of '27

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THESIS

Long-running agent populations can improve collectively through adaptive social dynamics and intelligence transfer.

Human innovation is an emergent property of long-running social dynamics; new ideas build on knowledge accumulated over generations through teaching, cooperation, and competition between groups.

- **Adaptive social dynamics:** have groups compete for finite resources, so that helping one's group is rewarded even if doing so costs an agent resources. Let agents revise how they cooperate, enter and leave swarms, and migrate between groups.
- **Intelligence transfer:** credit agents for the improvement of those they teach so that it is in their interest to spread useful knowledge; use teaching to combine discoveries from different lineages and have agents verify what they are taught before they adopt it.

RESEARCH

Our focus is swarm improvement itself.

Agent swarms have been observed to spontaneously develop social dynamics, but most existing swarms are one-shot, built for a short-horizon objective, or otherwise evolve independently and have a fixed search space. We let populations run continually, perturb organization and incentives, and measure how these affect what they discover, retain, and improve on, to identify what sustains collective improvement.

SANDBOXES

Sandbox 1: strategy development in skillful games

Skillful games provide clear credit assignment, counterfactuals, and metrics for agent interaction.

- Our swarms (xxx)
- developed a prior for the emergence of agent social dynamics and efficient organization under long time horizon games. DO THIS DO THIS DO THIS

Sandbox 2: autonomous research in market making

Market making lets swarms explore, exchange findings, compete for market share, and adapt to dynamic multivariate settings. We have domain knowledge to evaluate valuable problems and candidate strategies.

- 3–4 ms roundtrip execution with the Kalshi matching engine
- deployable positive expectancy strategies for market making